this variant (built on R-trees) uses spheres instead of rectangles.

Although it might seem a small change, there are strong theorical and practical evidences that suggest using spheres reduces the average number of leaves touched by a similarity (nearest-neighbor or region) query: we will discuss this point in more depth in section 10.5.1.

Each internal node `N` is therefore a sphere with a center and a radius; those two properties are uniquely and completely determined by `N`'s children: `N`'s center is, in fact, the centroid of `N`'s children[6], and the radius is the maximum distance between the centroid and `N`'s points.

To be fair, when we said that the only difference between R-trees and SS-trees was the shape of the bounding envelopes, we were guilty of omission: the choice of a different shape for the bounding envelopes also forces us to adopt a different splitting heuristic: in the case of SS-trees, instead of trying to reduce the spheres overlap on split, we aim at reducing the variance of each of the newly created nodes: therefore the original splitting heuristic chooses the dimension with the highest variance, and then splits the sorted list of children to reduce variance along that dimension (we'll see this in more details in the section about insertion, section 10.3.2).